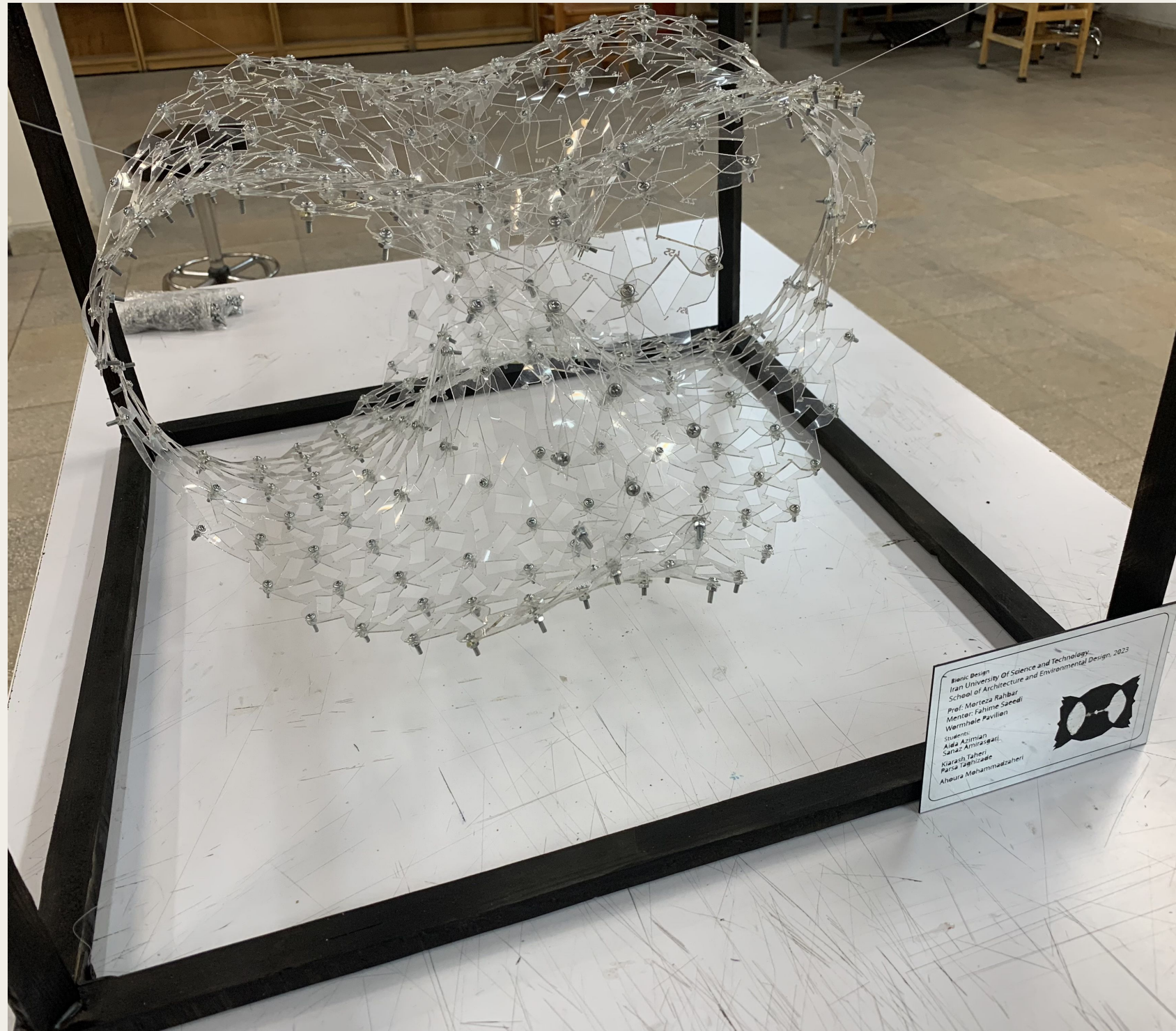


Wormhole

W04 saved spatial part field / computational design and team fabrication



A two-opening spatial field assembled from many small transparent parts. Photographs record bolted connections, suspension lines and the black display frame.

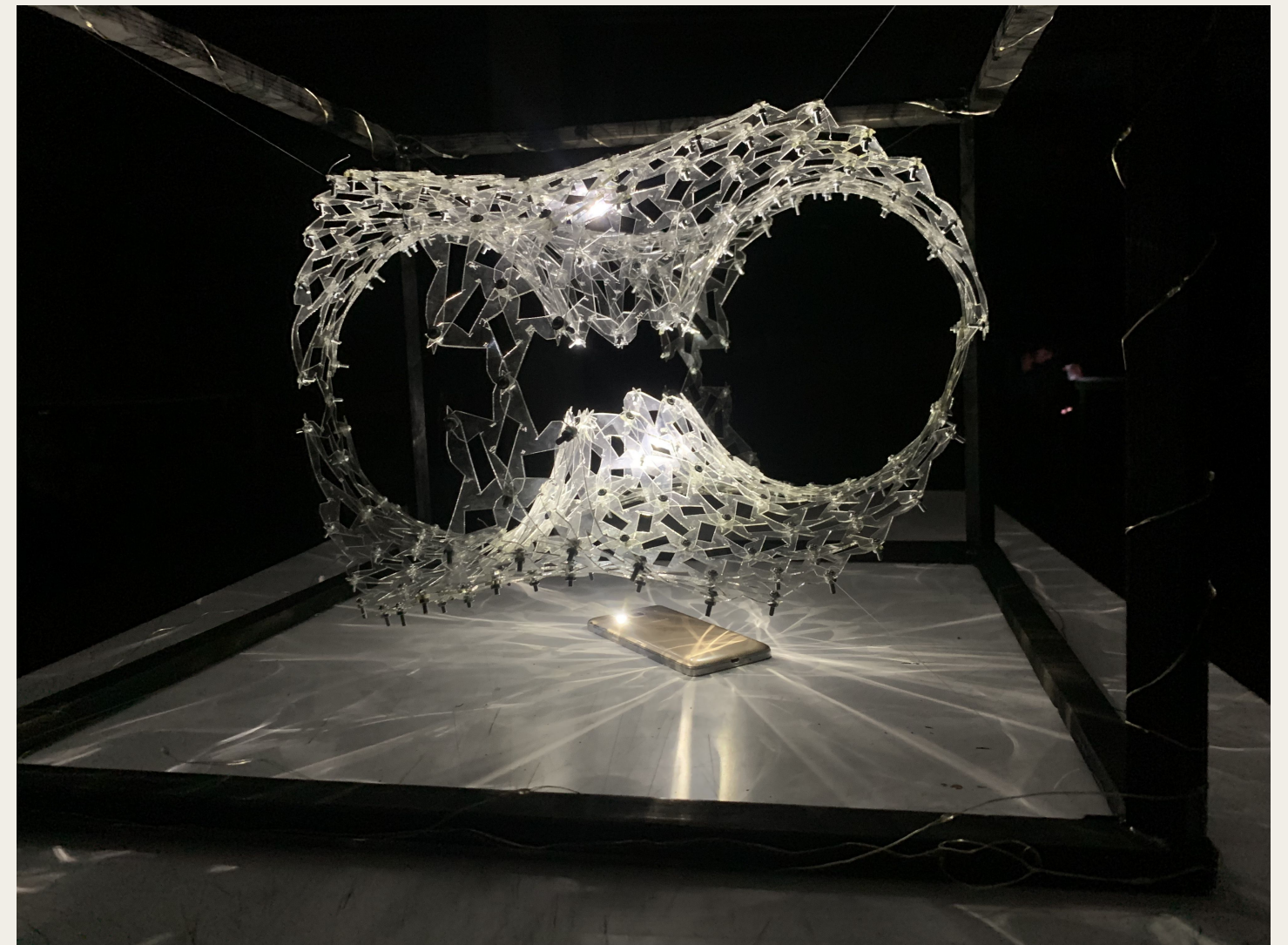
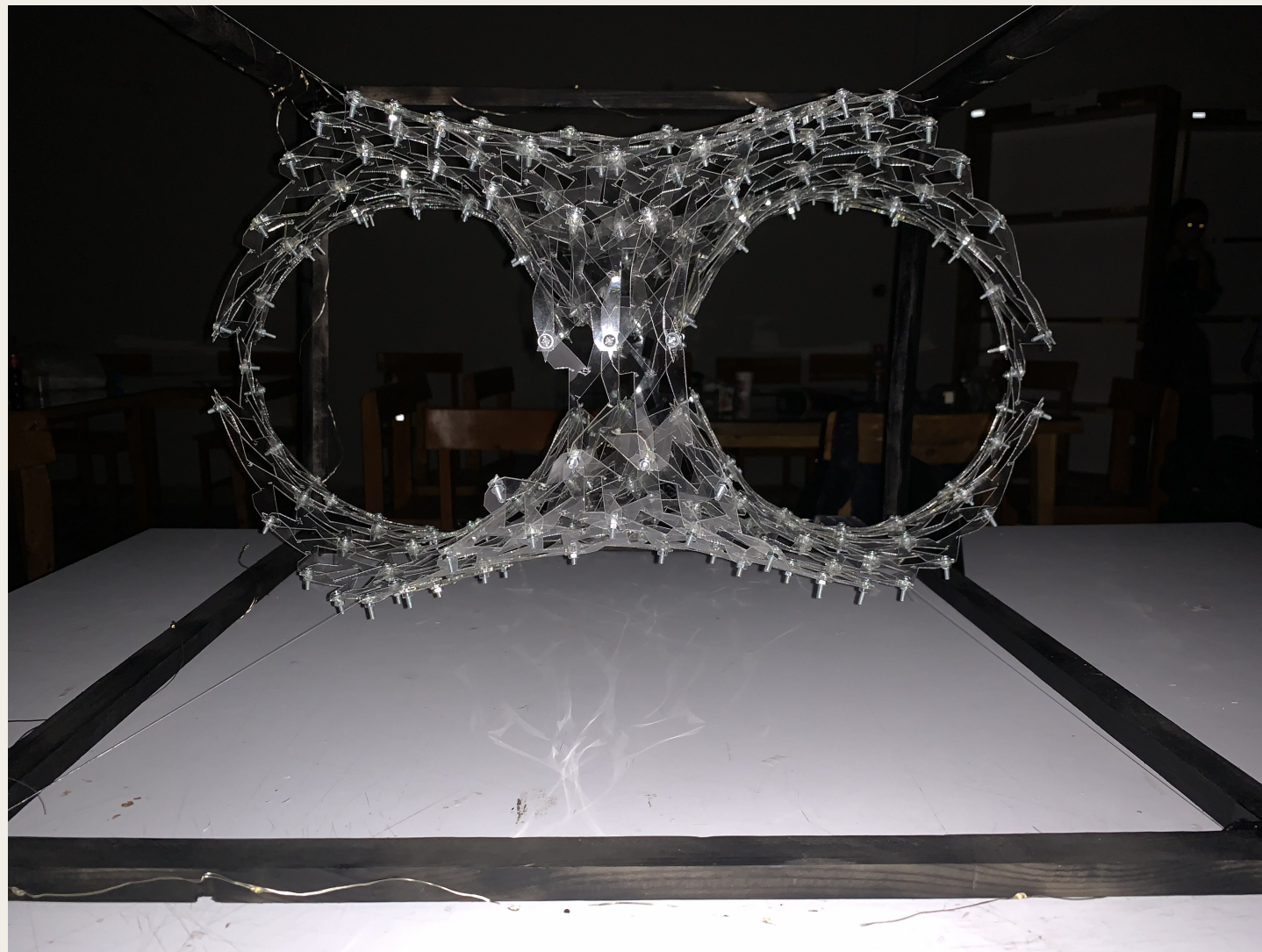
PROJECT CREDIT

Computational work: Kiarash Taheri.
Design: co-design with the team.
Fabrication: team participation.

The photographed project card identifies a 2023 Bionic Design studio project at Iran University of Science and Technology.

Read the built object

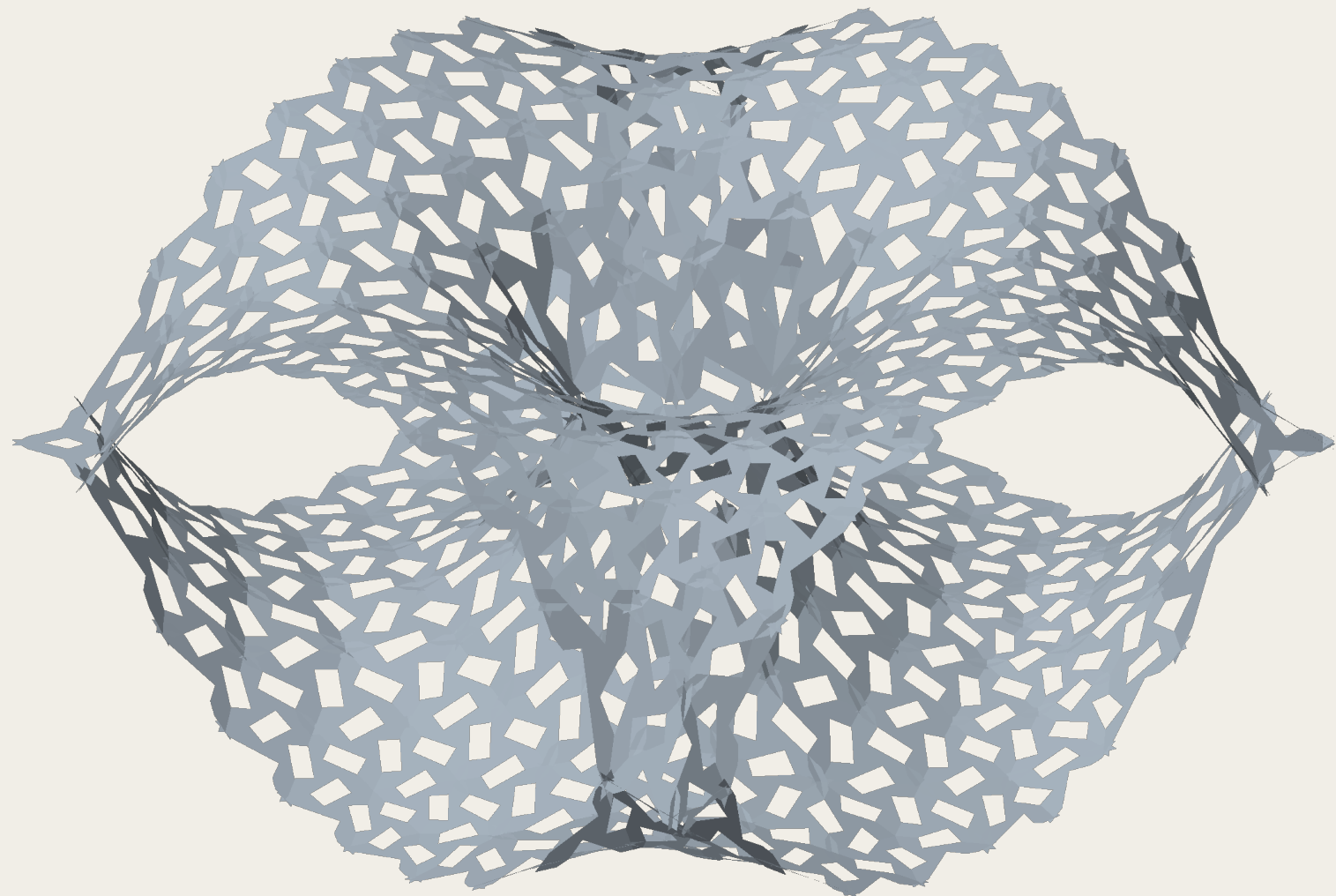
Documentary evidence establishes the visible assembly language



The frontal and oblique views show two openings joined through a central saddle, with a perforated field extending to the suspended boundary. The transparent surfaces and shadows make the connections and local orientations legible.

Recover the digital counterpart

442 original outline objects are isolated without changing their positions



W04 comes from the saved model shahkar.3dm. Its 442 closed polylines each contain 15 straight edges. The contours remain in their exact three-dimensional positions and retain their original object UUIDs.

New zero-thickness planar faces improve the display. They are bounded by the recovered contours; they do not add material, drill holes or fasteners.

The photographs and W04 belong to the same project context. An exact measured match to the manufactured revision has not been established.

Model-space dimensions

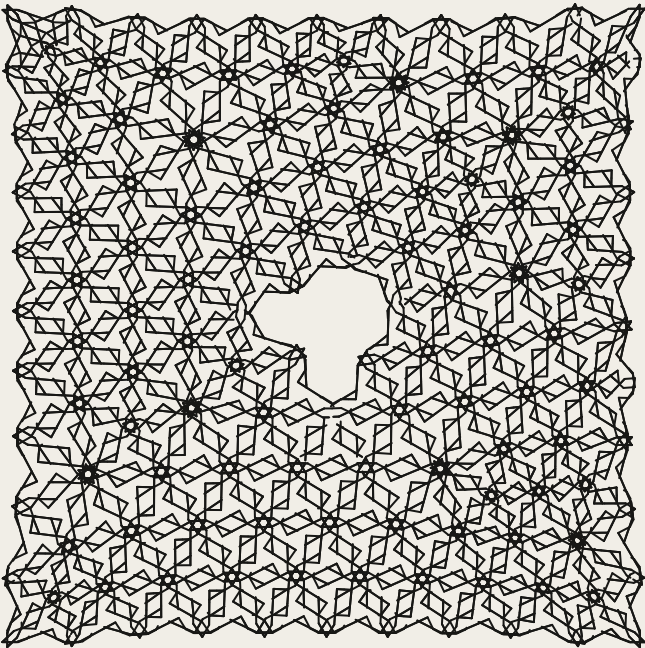
The drawing records source-coordinate extents

WH-02 / KIARASH TAHERI

Coordinate envelope

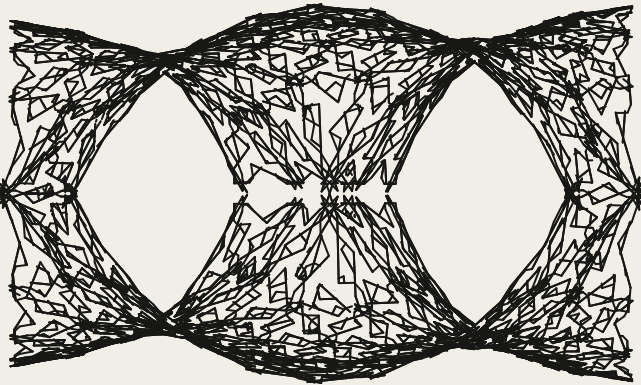
Saved-model extents converted from metres to millimetres

PLAN / X-Y



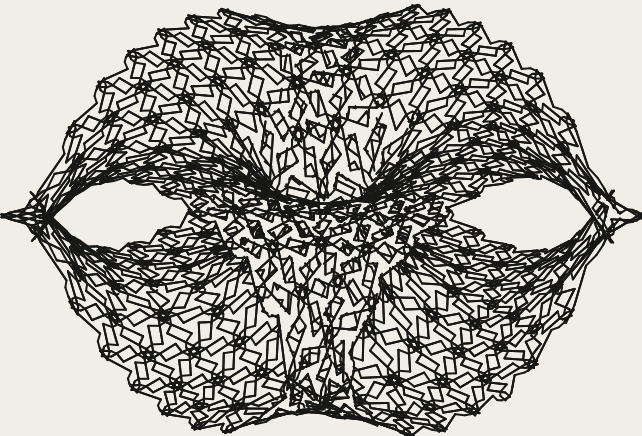
X 410.334 mm

ELEVATION / X-Z



Y 410.756 mm

AXONOMETRIC



Z 241.900 mm

These are source-coordinate dimensions, not a physical survey or a confirmed fabrication scale.

Planar parts, spatial composition

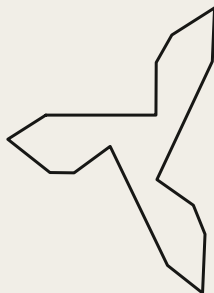
Every individual recovered contour passes the native Rhino planarity check

WH-03 / KIARASH TAHERI

From position to profile

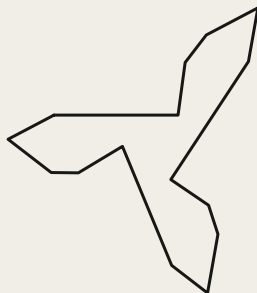
Eight recovered parts sampled across the area range

W0037



AREA 406.9 mm2

W0063



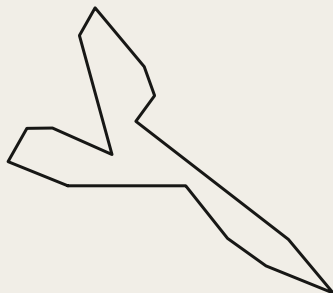
AREA 936.2 mm2

W0196



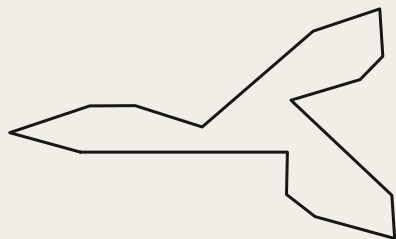
AREA 645.6 mm2

W0275



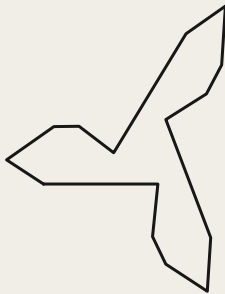
AREA 721.7 mm2

W0292



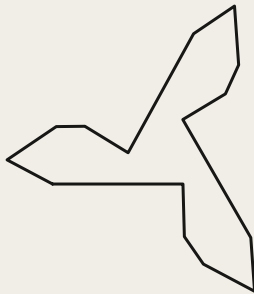
AREA 1394.1 mm2

W0391



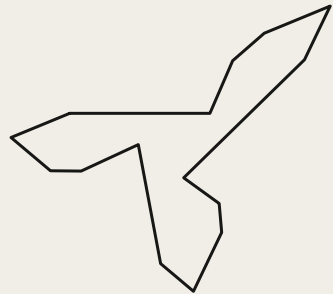
AREA 799.7 mm2

W0402



AREA 588.8 mm2

W0403



AREA 675.9 mm2

The earlier description of W04 as non-planar curves was incorrect. The field is spatial; its individual parts are planar. Each can be laid flat by a rigid orientation change.

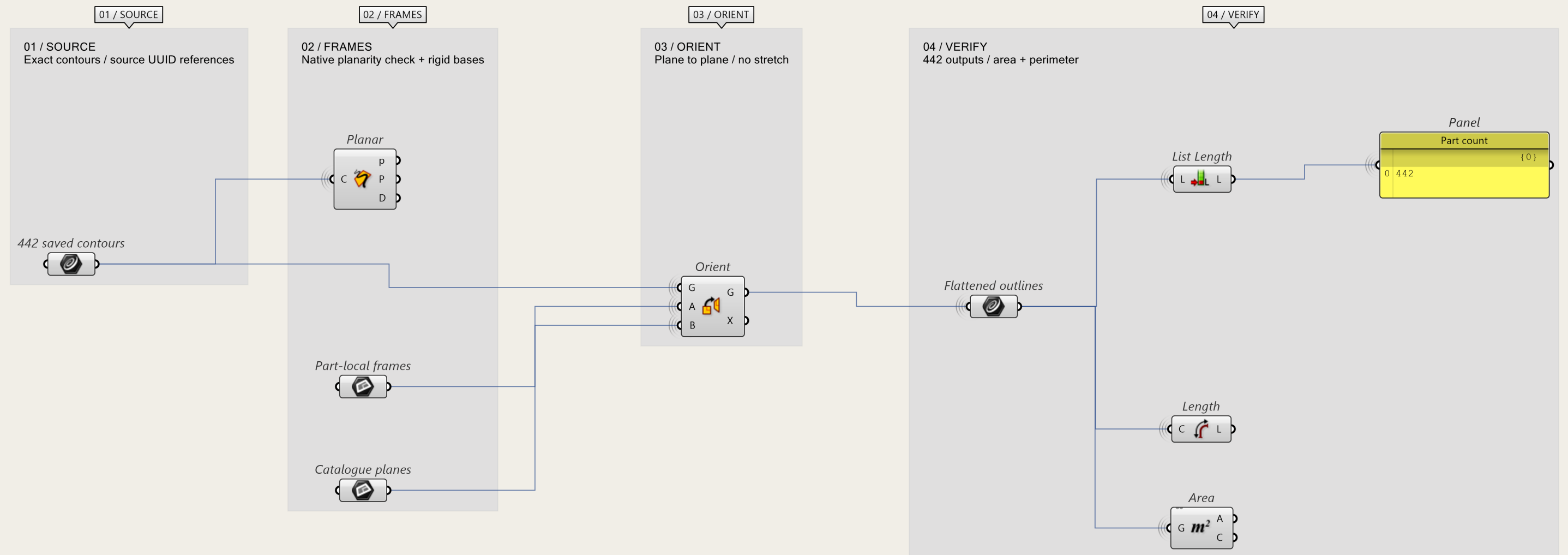
Area range: 406.9–1394.1 mm².
Perimeter range: 141.6–287.6 mm.
Planarity tolerance: 1e-9 m.

These measurements describe recovered outlines in source units. They do not establish stock thickness or the built model scale.

Every profile is rigidly oriented to its own plane; source UUIDs and original positions remain in the catalogue.

Rigid flattening in Grasshopper

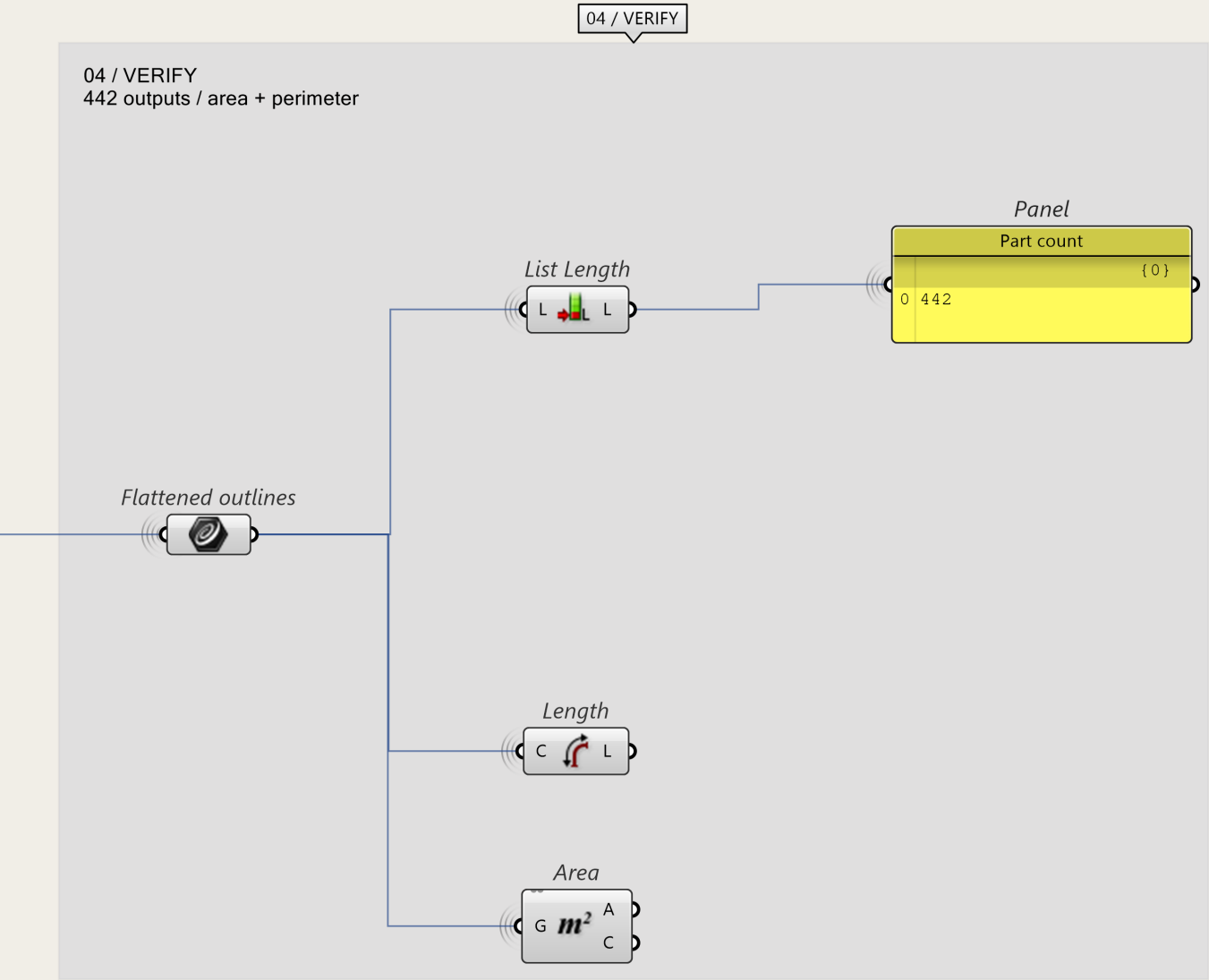
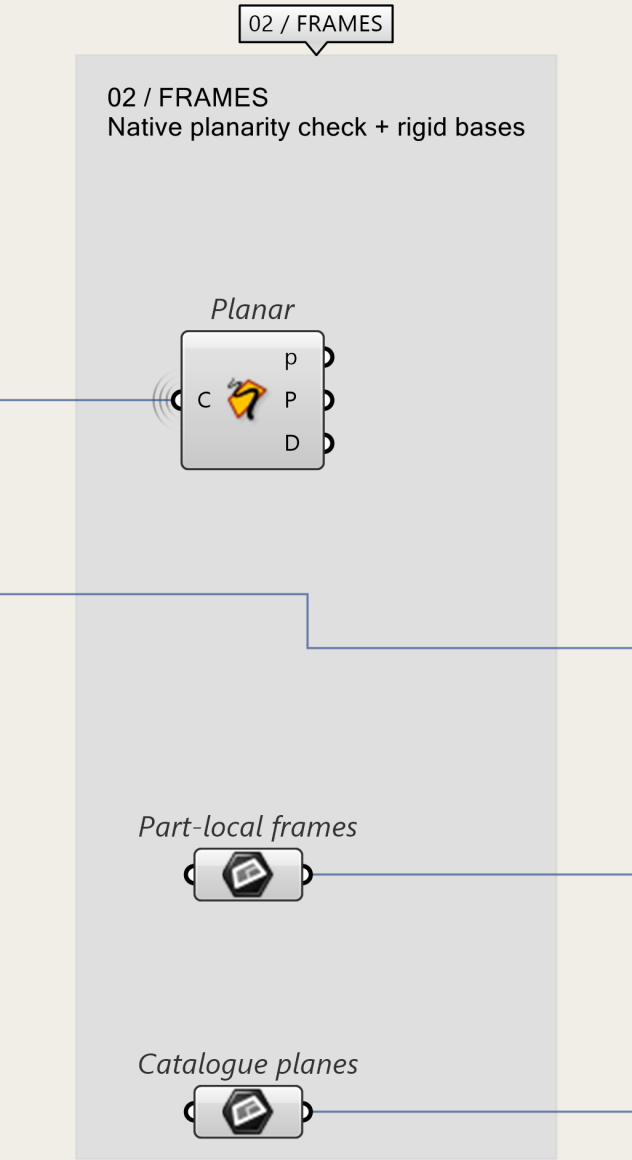
A new native inspection definition keeps the actual source model connected



Open W04-source-contours.3dm, then W04-native-flattening.gh. The definition references all 442 source UUIDs, checks their planes, orients each outline into its catalogue plane, and returns 442 flattened contours with perimeter and area outputs.

Frames and verification

Actual Grasshopper canvas close-ups / grouped native nodes and straight wire export



Local frames are computed from the retained contour coordinates. Catalogue target planes add only an organizational translation. Native outputs match the paired flat catalogue. The export draws straight segments through the native canvas event; saved definitions reopen with Grasshopper's standard wire appearance unless the supplied style helper is loaded.

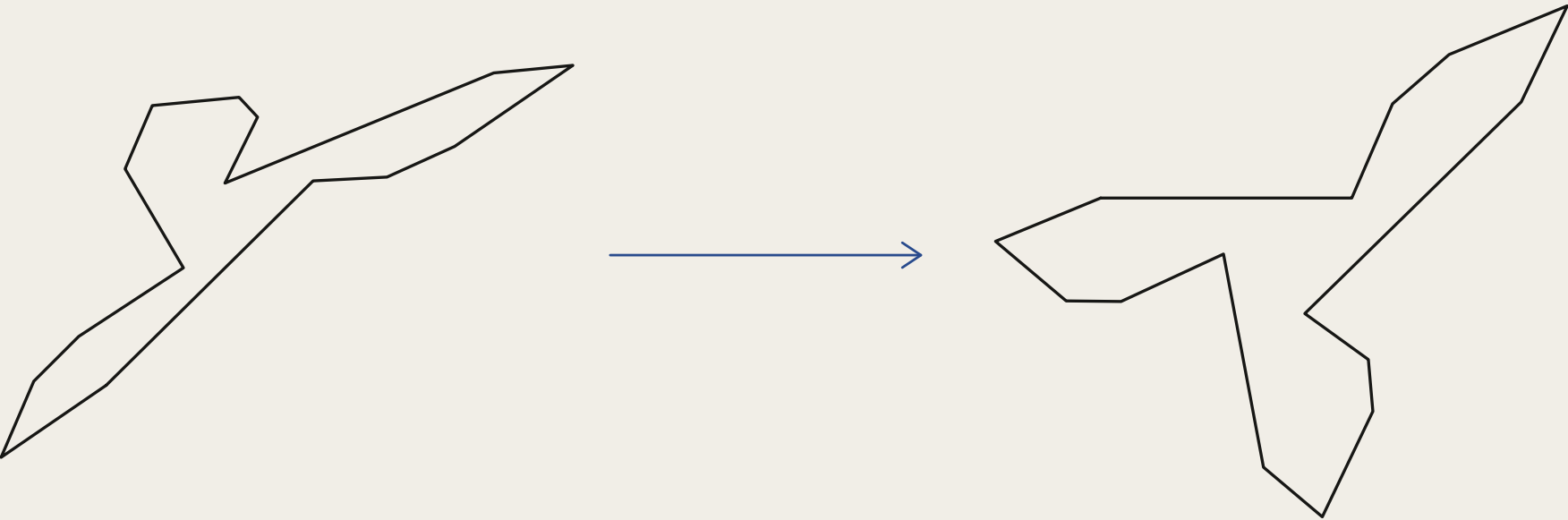
Part identity survives the drawing

A complete catalogue links every flat outline back to the spatial model

WH-04 / KIARASH TAHERI

A rigid flattening method

An orientation change preserves edge lengths and area



SOURCE PLANE

W0403 / SOURCE UUID
0dbaf981-3358-4e99-9bae-f755bcbedfd6

TARGET: WORLD XY

Native PlaneToPlane transform, with no stretch. Catalogue offsets organize the drawings and are not sheet nesting.

W0111

Source UUID:
8ef5affd-d641-4397-bc88-5c3cd1c28eae

The part index includes source position, local frame, perimeter, area and catalogue offset. W0001–W0442 are new recovery labels, not claimed original fabrication numbers.

The full catalogue contains every recovered profile. The millimetre DXF is an outline interchange file, with catalogue labels on a separate layer; it is not a production nesting solution.

Connection, material and assembly

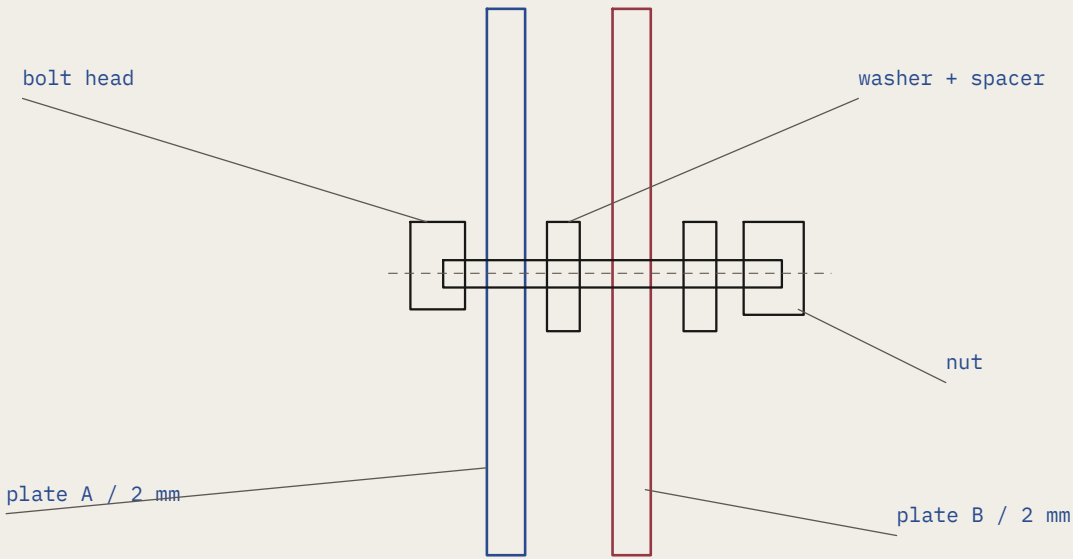
The photograph is the evidence; the detail is an explicitly new working study



WH-06 / KIARASH TAHERI

A working connection detail

Photograph-informed schematic / assumed clear acrylic, 2 mm sheet, M3 bolt



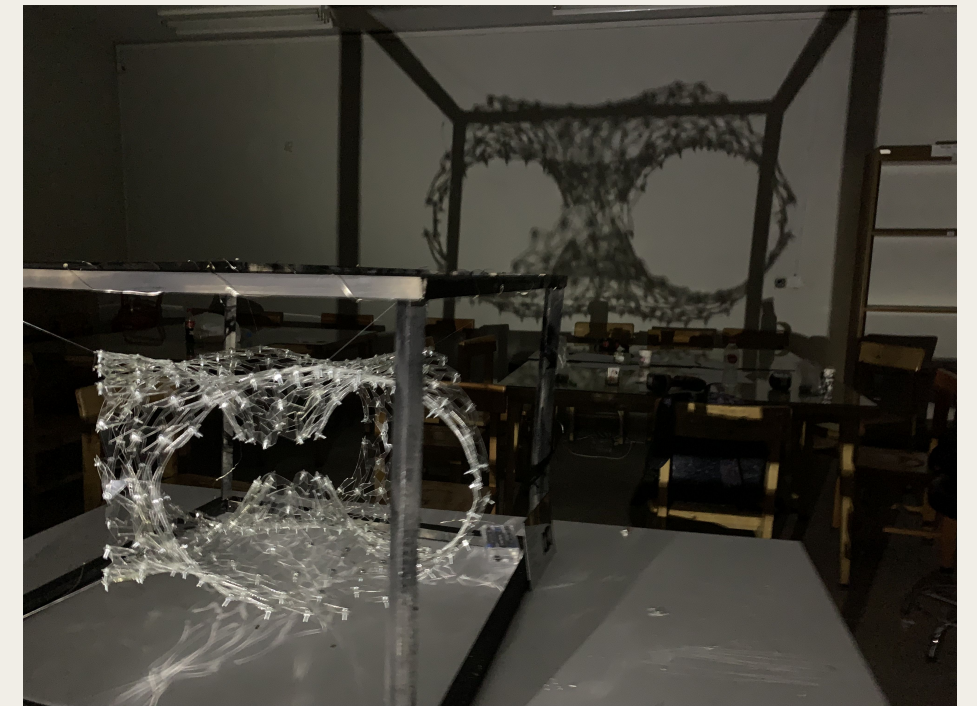
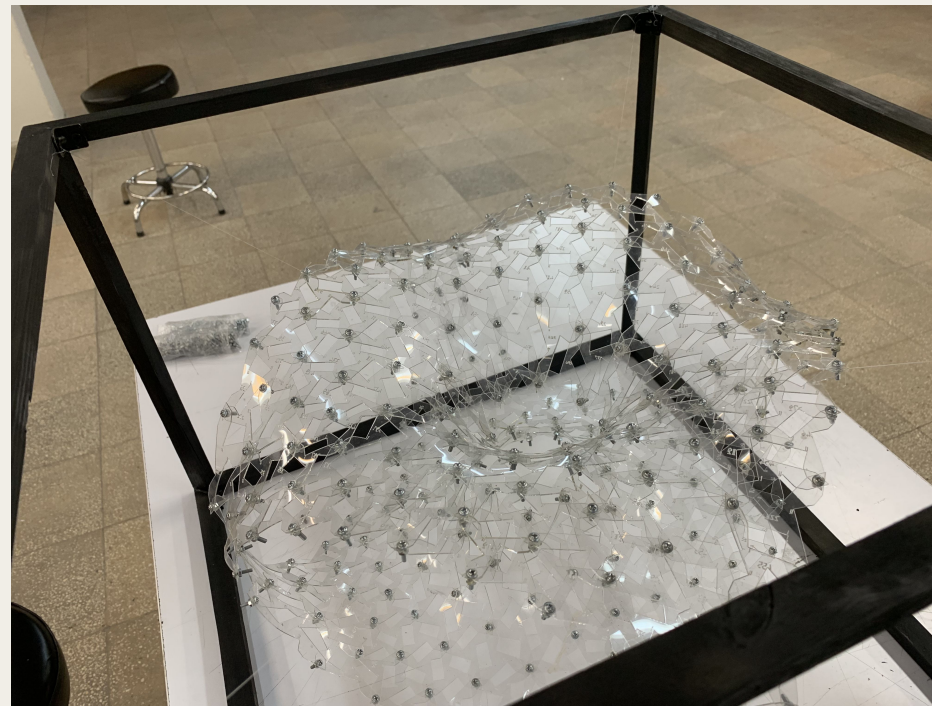
THICKNESS + FASTENER SIZE ARE WORKING ASSUMPTIONS

Schematic only; not a measured original joint or a fabrication-ready drill detail. Separation is exaggerated for clarity.

The photograph shows transparent sheet pieces with metal fasteners and washers. Clear acrylic, 2 mm sheet and an M3 bolt are sensible working assumptions for the explanatory detail, authorized by the project author; they are not recovered specifications.

Assembly as a spatial operation

A proposed reading sequence from the actual photographs



01 / local orientation

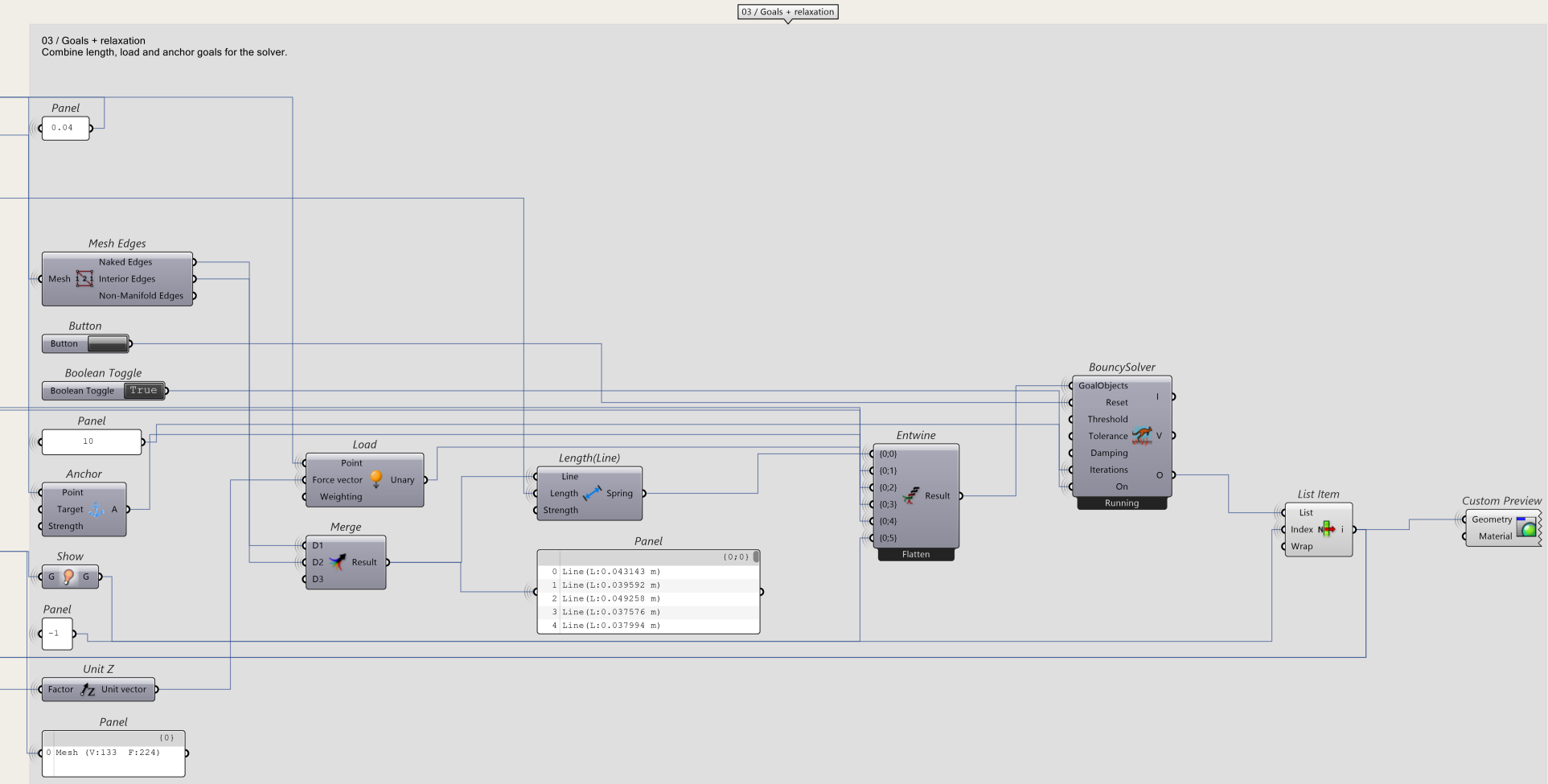
02 / boundary suspension

03 / field + shadow

For a new demonstration, identify the contours and local joints first, build connected patches second, and reconcile the upper boundary with suspension last. This is a proposed explanatory sequence; the recovered photographs do not document a complete historical assembly order.

The historical computational chain

Saved surface controls, relaxation goals, triangle mapping and nesting



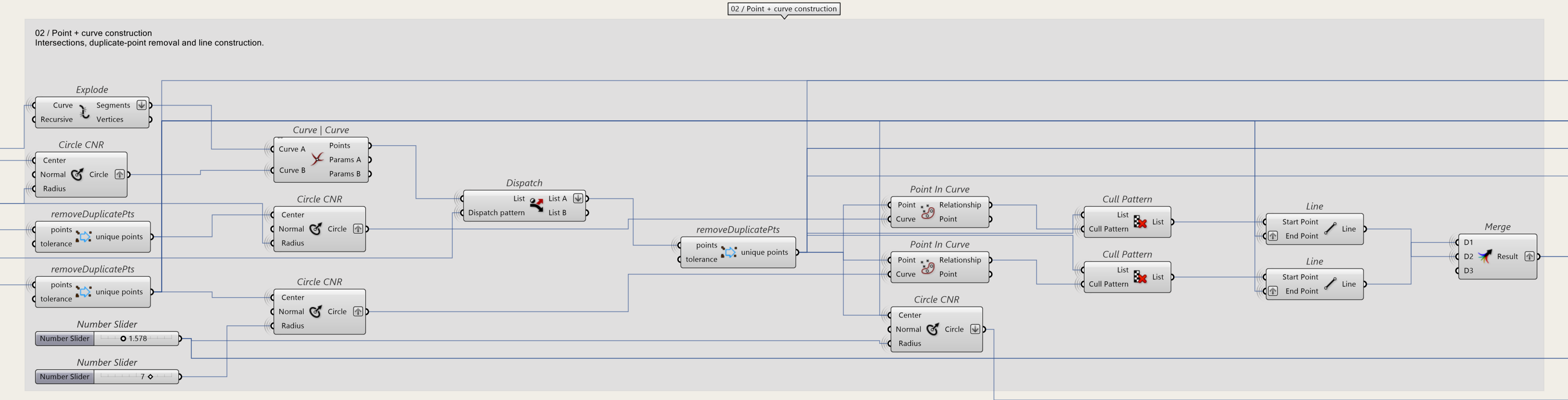
The original 0069 definition is retained as a historical algorithm. Its actual nodes and connections are grouped into input, boundary, relaxation, mapping and labelling/nesting stages. Native recalculation uses UUID-matched inputs.

The saved solver parameters and wire logic are unchanged. A 2026 recalculation provides a reproducible snapshot; it does not prove dynamic convergence or that the output is the manufactured W04 revision.

Full canvas and all five stage images are supplied at native high resolution.

How a triangle becomes a part

Historical local-part construction, with the actual matching triangle input



The 0073 definition records local part construction. The matching triangle model and native definition are included together. This is evidence of the computational method; the W04 contour catalogue is a separate recovery from the saved spatial field.

Cutting evidence and recovery limits

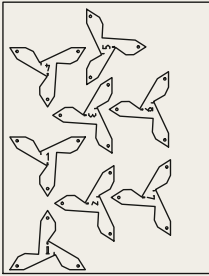
Retained fabrication layouts are distinct from the new W04 outline catalogue

W-02 / KIARASH TAHERI

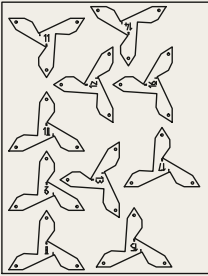
Nine cutting layouts

The retained sheets, recomposed into a readable array.

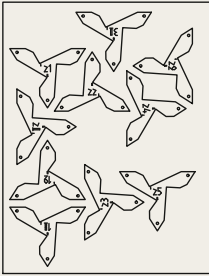
01 / 8 PARTS



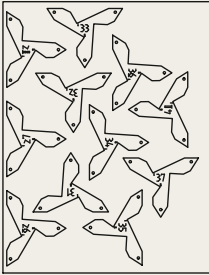
02 / 10 PARTS



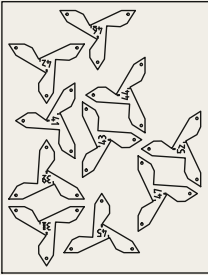
03 / 10 PARTS



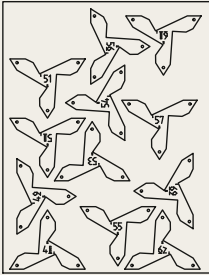
04 / 11 PARTS



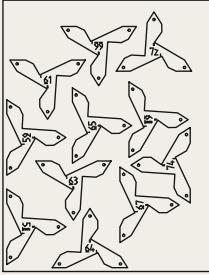
05 / 10 PARTS



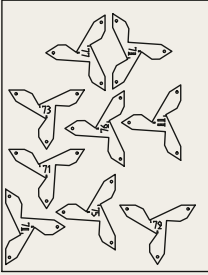
06 / 12 PARTS



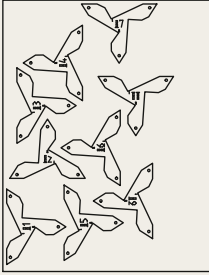
07 / 11 PARTS



08 / 9 PARTS



09 / 9 PARTS



9 source sheet frames / 90 inserted parts. Individual geometry is unchanged; sheet positions are recomposed for reading.

The archive also contains a retained cutting DXF and mapped/nested native outputs. Keep these source branches distinct: shared project context does not establish that every cut profile is the same revision as W04.

For a new fabrication run, measure the actual sheet stock, choose joint clearances, recover or author drill features, set kerf and stock sizes, and then nest. Those decisions are intentionally absent from the outline recovery.

A usable technical handoff

Editable geometry, native computation, reusable artwork and documentary originals

Model

W04-source-contours.3dm retains 442 source curves and UUIDs. W04-planar-profile-faces.3dm supplies new display faces. The flat catalogue model and millimetre DXF retain outline identity.

Computation

W04-native-flattening.gh is the new inspection definition. Historical 0069 and 0073 definitions have matching input models, unchanged logic, full native canvas exports and stage close-ups.

Artwork

WH-01–WH-07 are supplied as outlined-text SVG, transparent PNG, artwork-only SVG and editable drawing data. CAD display PNGs and GLBs come from native tessellation.

Documentation

The full 442-part catalogue, CSV and JSON carry source UUIDs and measurements. Eight documentary photo masters are unchanged. Project credit and material assumptions are explicit.